

Leonardo Herrera

Cell Phone: (281) 900-6047

Spring, TX

Portfolio: herrera2.github.io/Personal-Website/index.html

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LinkedIn: linkedin.com/in/lh-mech-eng

Skills

SolidWorks, Creo Parametric, Siemens NX, AutoCAD, GD&T, Tolerance Stack-Up, MATLAB, System Identification, Transfer Functions, Simulink, FEA, System Identification, Hardware-in-the-Loop, Systems Integration, Sensor Calibration, CNC Nesting, Sheet Metal Design, Root Cause Analysis, Process Improvement, Trilingual (English, Spanish, Portuguese)

Work Experience

Manufacturing Design Engineer

May 2025 – Present

Duke Manufacturing – CounterCraft

Houston, TX

- Design and release structural sheet-metal components in SolidWorks, supporting full product lifecycle from concept through production for custom-engineered systems.
- Develop flat patterns, bend allowances, and CNC nesting strategies, improving material efficiency and reducing waste. Perform tolerance stack-up analysis to ensure alignment and fit across multi-component systems.
- Implemented a parametric universal mullion design framework in SolidWorks to standardize internal support structures across custom countertop assemblies, reducing modeling time by ~15-20%.
- Partner directly with fabrication and shop-floor teams to troubleshoot structural nonconformances, perform root cause investigations, and implement corrective design or process improvements.

Mechanical Design & Test Engineering Intern

March 2022 – September 2023

NASA Johnson Space Center – CACI (Contractor), PVAMU (Subcontractor)

Houston, TX

- Supported spacecraft docking certification campaigns comprising 300+ test cases, contributing to setup, calibration, and execution of dynamic hardware-in-the-loop simulations.
- Integrated and configured test articles and instrumentation (electromechanical instrumentation, optical tracking, laser metrology) for complex multi-system test operations. Supported test planning throughout full test cycle.
- Operated a 6DOF robotic system in closed-loop simulations, where real-time sensor feedback informed physics-based models to dynamically adjust motion during docking events. Interpreted sensor and system data during test execution.
- Designed and improved test hardware (sensor mounts, structural interfaces) using Creo Parametric and Creo Simulate, increasing measurement reliability and system integration performance.
- Developed calibration tooling and validated force measurement systems, improving data accuracy and repeatability.
- Built dynamic system models in MATLAB/Simulink, applying system identification and transfer function analysis to characterize system behavior.
- Collaborated with multidisciplinary engineering teams to support mission-critical validation efforts for partner vehicles including SpaceX and JAXA.

Technical Projects

Robotic Arm Simulation (NASA Trick) (*In Development*)

May 2026 – Present

- Developing 3-DOF robotic arm simulation in NASA Trick (Python/C++); implementing forward kinematics and inverse kinematics framework for real-time control and visualization

Stewart Platform Kinematics & Simulation (stewart_platform.html)

March 2026 – April 2026

- Developed inverse/forward kinematics for a 6-DOF Stewart platform; solved constrained actuator lengths via numerical optimization and enabled real-time 3D motion visualization

Education

Master of Science in Engineering

Graduated December 2023

Prairie View A&M University, Prairie View, TX

GPA: 3.80

- Thesis: Control Filter Assessment on Six Degree of Freedom Stewart Platform (Publication Pending)

Bachelor of Science in Mechanical Engineering

Graduated May 2022

Prairie View A&M University, Prairie View, TX

GPA: 3.67

- Senior Project: Designed, modeled, and built a prototype of a Lightweight Deployable Shield for Ball Aerospace.

Awards

Elite Team Award, NASA

September 2023

Elite Team Award, NASA

August 2022